

ಆದಿಕವಿ ಮಹರ್ಷಿ
ವಾಲ್ಮೀಕಿ
ವಿಶ್ವವಿದ್ಯಾಲಯ
ರಾಯಚೂರು



ADIKAVI
MAHARSHI
VALMIKI
UNIVERSITY
RAICHUR

PROJECT REPORT

ON

“CRIME SCENES DETECTION USING DEEP LEARNING”

Submitted In Partial Fulfillment of The Requirements

For

VI SEMESTER

BACHELOR OF COMPUTER APPLICATIONS

SUBMITTED BY

SHUSHIL KUMAR PULSE

U23IA22S0089

Under The Guidance Of

Prof. MANJUNATH. V B. E, M.Tech, NET

Assistant Professor



DEPARTMENT OF COMPUTER APPLICATIONS

GOVERNMENT DEGREE COLLEGE,

YADGIR – 585202

2024 – 25

ಆದಿಕವಿ ಮಹರ್ಷಿ
ವಾಲ್ಮೀಕಿ
ವಿಶ್ವವಿದ್ಯಾಲಯ
ರಾಯಚೂರು



ADIKAVI
MAHARSHI
VALMIKI
UNIVERSITY
RAICHUR



GOVERNMENT DEGREE COLLEGE, YADGIR
DEPARTMENT OF COMPUTER APPLICATIONS

CERTIFICATE

This is to certify that the project entitled “**Crime Scenes Detection Using Deep Learning**” has been successfully submitted by **SHUSHIL KUMAR PULSE** RegNo. **U23IA22S0089**.

A Bonafide student at Government Degree College, Yadgir in partial fulfillment of award of Bachelor of Computer Applications in the year 2024 – 2025 is record of student own work carried out under my/our guidance. It is certified that all corrections/suggestion indicated for internal assessment have been incorporated in the report & one copy of it being deposited in the Government Post-Graduation Library. The project report has been approved as satisfies the academic requirements in respect of project work prescribed for the said bachelor’s degree. It is further understood that this certified the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion draw there in but approved the project only for the purpose for which it is submitted.

Examiners:-

1.

2.

Guide
Name & Guide signature

HOD
Department of Computer Applications

ACKNOWLEDGEMENT

I express my sincere gratitude to my guide, **Shri. Manjunath. V, Assistant Professor** of department of computer applications to guide for their guidance, proper advice & encouragement during the course or work.

I also feel very obliged to **Dr. Basavaprasad. B, Head of the Department** of computer applications for his encouragement & inspiration for execution of this thesis work.

I also feel thankful to **Dr. K Subhashchandra, Principal** of Government Degree College, Yadgir Support during work.

I am also thankful to the entire faculty & staff members and my friends from the computer applications department for their direct & indirect help & cooperation.

Date:-

SHUSHIL KUMAR PULSE

Regno:- U23IA22S0089

CANDIDATE DECLARATION

SHUSHIL KUMAR PULSE a student of Bachelor's in Computer Applications department bearing Regno **U23IA22S0089** of **Government Degree College, Yadgir** hereby declare that my full own responsibility for the information, results and conclusions provided in this work titled "**Crime Scenes Detection Using Deep Learning**" submitted to **Adikavi Maharishi Valmiki University, Raichur** for the award of Bachelor's Degree In Computer Applications.

To the best of my knowledge, this project has not been submitted in part or full elsewhere in any other institution/organization for the award of any certificate/diploma/degree. I have completely taken care of acknowledging the contribution of others in academic work. I further declare that in any case the candidate will be solely responsible for the same.

Date:-

Place:-

Signature of the candidate

Name:- SHUSHIL KUMAR PULSE

Regno:- U23IA22S0089

Abstract

In recent years, the increasing prevalence of criminal activities has created a significant need for intelligent surveillance systems that can proactively monitor environments and detect crime in real time. Traditional surveillance methods heavily rely on manual monitoring, which is not only labor-intensive but also prone to human oversight and delays in response. To address this issue, this project proposes an AI-powered video surveillance solution that leverages deep learning techniques to automatically detect and classify crime scenes.

The system takes video input—either from a file or a real-time camera feed—and processes it to identify one of four predefined crime categories: **Arrest, Abuse, Arson, and Assault**. To achieve this, the project employs a hybrid deep learning model that combines **Convolutional Neural Networks (CNNs)** for spatial feature extraction with **Long Short-Term Memory (LSTM)** networks for capturing temporal dynamics across video frames. This combination enables the system to learn complex motion patterns and contextual cues associated with different types of criminal activities.

The model is trained on a labeled dataset containing video clips of various crime scenarios, which are preprocessed through frame extraction, resizing, and normalization. During inference, the system analyzes sequences of frames, classifies the activity based on learned features, and overlays the detected crime type on the video feed in real time. The implementation also includes a user-friendly interface that allows easy visualization of predictions.

Experimental results show that the model achieves high accuracy in classifying crimes, with robust performance even in low-light or cluttered scenes. The use of LSTM layers significantly enhances temporal context understanding, which is crucial for distinguishing between visually similar actions that occur over time.

This project demonstrates how deep learning can be effectively used for intelligent video analysis in public safety applications. With further enhancements such as real-time alert systems, extended crime categories, and deployment on edge devices or cloud platforms, this system has the potential to revolutionize modern surveillance infrastructure. Its applications span across law enforcement, smart city development, and private security systems, where timely and accurate crime detection is critical for proactive intervention and safety assurance.

Keywords: -

Crime Scene Detection, Deep Learning, Video Classification, Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), Real-Time Surveillance, Crime Recognition, Intelligent Video Analysis, Activity Recognition, Public Safety AI, Temporal Feature Extraction, Computer Vision, Automated Monitoring, Smart Surveillance, AI in Law Enforcement.

CONTENTS

Description	Page No.
1. INTRODUCTION	1
1.1 Machine Learning	2
1.2 Deep Learning	2
1.3 Objective	3
1.4 Problem Statement	4
2. LITERATURE SURVEY	6
3. SYSTEM ANALYSIS	9
3.1 Existing System	9
3.2 Proposed System	10
3.2.1 Deep Learning Approach	12
4. ALGORITHMS	15
4.1 3D Convolutional Neural Networks (3D CNN)	15
4.2 3D Max Pooling	16
4.3 Softmax Classification	17
4.4 Cross-Entropy Loss	17
4.5 Random Clip Sampling from Video	18
4.6 Adam Optimizer	19
5. REQUIREMENTS SPECIFICATIONS	21
5.1 Hardware Requirements	21
5.2 Software Requirements	21
5.3 Libraries File Requirements	22
5.3.1 OpenCv	22
5.3.2 Numpy	22
5.3.3 Pytorch	23
5.3.4 Torchvision	23
5.3.5 TQDM	24
5.3.6 OS	25
5.3.7 Streamlit	26
6. IMPLEMENTATION	27
7. RESULTS	30
8. CONCLUSION & FUTURE SCOPE	31
REFERENCES	33